Commissioning Manual

Trimble® Earthworks Landfill Compactor

For SITECH Only

Version 2.24.x
Revision A
March 2026

TRANSFORMING THE WAY THE WORLD WORKS

Contents

1

Landfill compactor commissioning checklists

In this guide:

1.1 Checklist—Configuring a system

- ☐ Connect all system components:
  - EC520
  - Display
  - GNSS receivers
  - Remote switch (optional)
  - Internet gateway (optional)
  - Radio (optional)
- ☐ Complete the EC520 firmware installation as per the release notes.
- ☐ Turn on the ignition to start the display.
- ☐ Confirm that the display's date and time are correct. To change the date and time, open Settings > System > Date & Time > Time Zone and select the correct time zone.
- ☐ Tap the Web Interface icon and log in.
- ☐ If this is the first Web Interface log in:
  - set an admin password
  - configure Wi-Fi
  - complete the setup section of the Install Assistant
- ☐ If this is a subsequent Web Interface login – the home page opens.
- ☐ Open Advanced > Licenses and apply a core bundle license.
- ☐ Open Configure > Install Assistant and complete each section. Use a TD5x0 or Wi-Fi BYOD display to perform valve calibrations.
- ☐ Select Network > GNSS Correction Source and select a correction source.
- ☐ Select Monitor > GNSS Details and check the GNSS receiver.
- ☐ Close the Web Interface, open the Operator App, log in and log out again. If the Operator App launcher is not installed, use MC Installer to install it.
- ☐ Open the Web Interface's Advanced > Licenses screen and apply the remaining module licenses to the display.
- ☐ If you have a WorksManager/Connected Community subscription:
  - Select Network > Cloud Services; set up WorksManager/Connected Community.
  - Upload project files to WorksManager/Connected Community to sync to the machine.
  - Select Operation > Projects and set up the interval the EC520 synchronizes with WorksManager/Connected Community.
- ☐ Close the Web Interface again, and re-open the Operator App.
- ☐ If you are using a USB flash drive to transfer project files onto the system, insert and transfer from the USB flash drive to the machine.
- ☐ Confirm that all tiles on the Dashboard are not red. The Start button is then enabled, and tapping the button opens the work screen.
- ☐ Touch and hold on the text ribbon to configure the text items.

- ☐ Confirm that 3D guidance is correct: confirm the Northing, Easting and Elevation position at the Focus tip.

  Test the system gives accurate position while working on a design.

2

Landfill compactor measure-up best practices guide

In this guide:

2.1 Overview of landfill compactor measure-up

2.1.1 Purpose of this guide

This guide supports the precision measure-up of the compactor body and guidance components as part of the configuration of a Trimble® Earthworks compactor system.

The measure-up procedure generates the measured values entered into the Measure-up section of the Web Interface's Configure > Install Assistant screen. Sensor data is taken while the Measure-up section screens are completed, so it is critical that the machine is in the measure-up position both when the points are measured ***and*** when the values are entered into the Measure-up section.

The measure-up procedure requires the use of a total station instrument to achieve the measurement accuracies needed for accurate positioning.

2.1.2 Prerequisites

Before starting the compactor measure-up:

- Know how to set up and use a total station, for example a Trimble TS/SPS Series total station or a Nikon M Series total station.
- Know how to use SCS900/Siteworks Site Controller software, or the total station's on-board software.
- Install the cab and body components, and validate the installation.
- Complete the preceding sections of the Web Interface's Configure > Install Assistant screen.

- Review this entire guide. This ensures that you understand the full measure-up workflow.

2.1.3 Measure-up workflow

1. Prepare for the measure-up:
    a. Position the machine.
    b. Mark all measurement points.
    c. Place the measure-up targets on the marked points.
    d. Position the total station.
2. Measure the position of all points.
3. Complete the Measure-up section of the Install Assistant to enter the point positions and take the sensor readings.
4. Remove the measure-up targets.

2.1.4 Equipment list

- Total station, in good adjustment, and with a charged battery
- If required, a data collector with a charged battery, and software
- Measure-up target kit (P/N 105568)
- Cleaner/de-greaser
- Cleaning rags
- Marker pen – choose a color easily visible against the frame of the compactor
- Self-adhesive or magnetic acrylic targets (do not use DR measurement mode, it does not provide the accuracy of measurement required)
- Scissors
- Straight edge
- Digital level
- Tape measure
- These instructions

2.1.5 Machine setup

Before proceeding with the measure-up process, ensure the EC520 is mounted per the installation instructions. If it is mounted in a different position or orientation, the accuracy of the system will be compromised.

If the EC520 is moved or replaced at any time, you must perform the Measure-up section of the Install Assistant again.

2.2 Preparing for measure-up

2.2.1 Overview

A precision measure-up requires:

- A stable, flat pad for the machine
- A stable, safe position for the total station instrument

2.2.2 Set up the machine for measurement

Machine position

Park the machine on a level, smooth, hard surface.

Drum position

Ensure there is no articulation, so the rear drum is approximately square to the machine.

2.3 Mark measure-up points

2.3.1 Introduction

This section describes how and where to attach the acrylic targets for the total station to measure.

> **NOTE –**Do not use DR measurement mode. Always measure to acrylic targets to get an accurate measure-up.

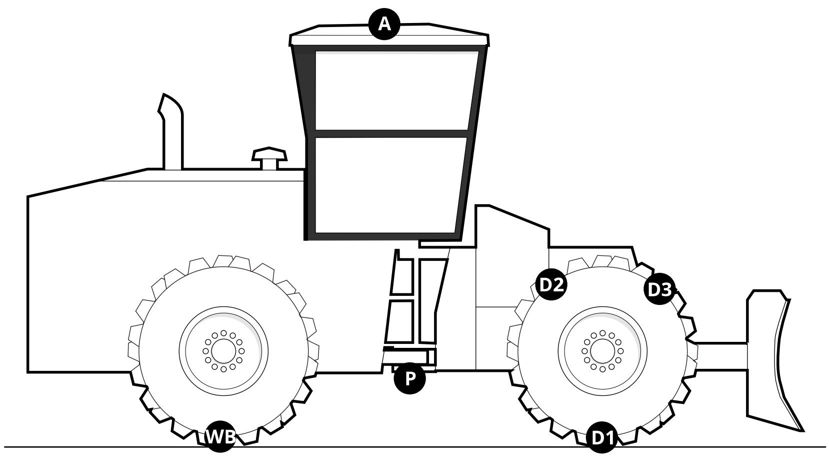
2.3.2 Best practices

> **NOTE –** Position targets accurately. Your measurements cannot be more accurate than your target positions.

- *Clean* all measurement points.
- *Mark* all measurement points with an acrylic target.
- *Label* all measurement points with the point ID. Write the ID beside the target so you can see it through the instrument.

2.3.3 Mark measurement points

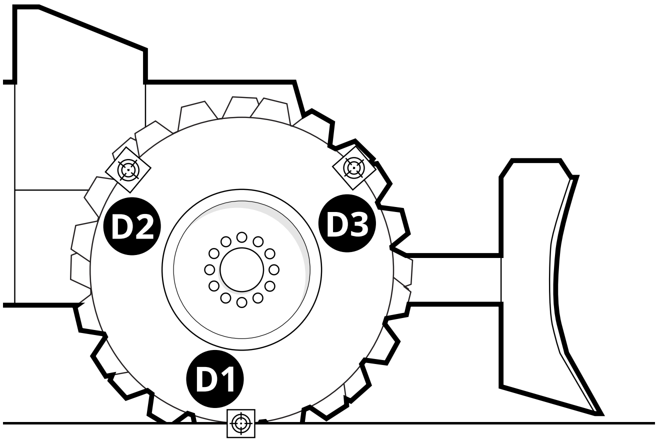
This section shows how to place the targets to perform a compactor measure-up.



Drum (D1, D2 and D3)

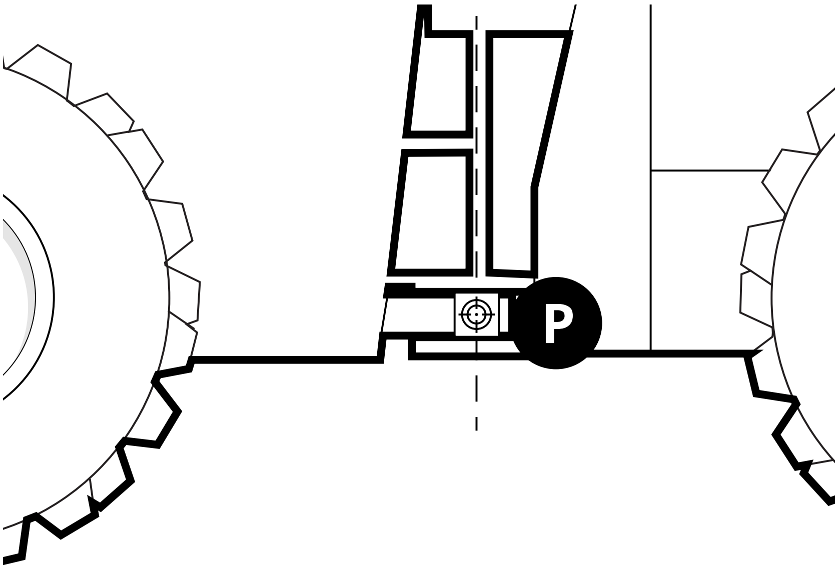
1. Place an acrylic target on the rim of the drum at its lowest point so the target crosshairs mark where the drum pad meets the ground. This is point D1.
2. Place 2 more targets on the drum rim, at approximately evenly spaced distances. Point D2 is behind the drum axle and point D3 is in front of the drum axle.

Ensure that the 3 points are all on the same plane. Together, these 3 points enable the system to calculate the position and diameter of the drum.



Pivot point (P)

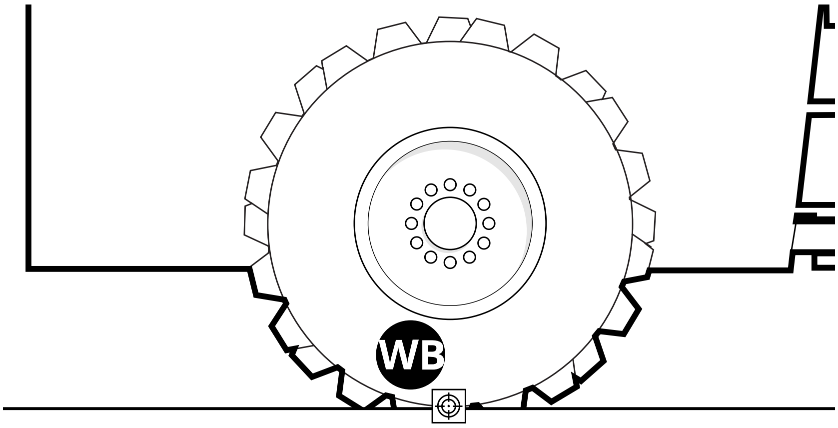
Place a target (point P) on the pivot point. Ensure the target crosshairs mark the axis of articulation in the machine's forward/back direction. It is acceptable for the target to be closer to the total station than the axis of articulation.



Base of rear drum (WB)

Place a target (point WB) on the lowest point on the rear drum so the crosshairs mark where the drum contacts the ground.

The accurate positioning of targets WB and D1 is important to ensure the body sensor mainfall is accurately calibrated.

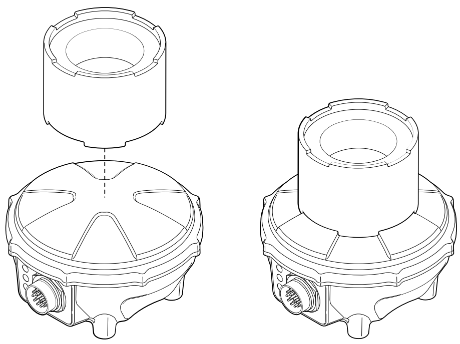


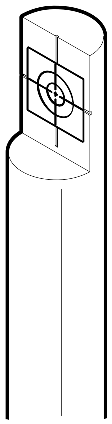
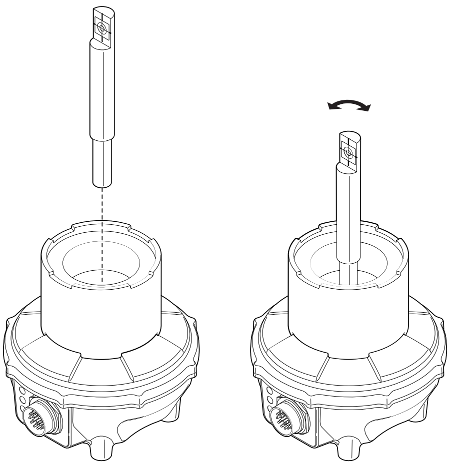
GNSS receiver/antenna position (A)

Use a measure-up target kit (105568) to measure the position of the cab-mounted GNSS receivers or antennas.

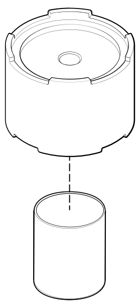
In the Setup section, you can select whether the target is sitting on a GNSS receiver or on a quick-release mast.

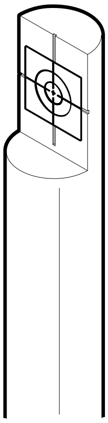
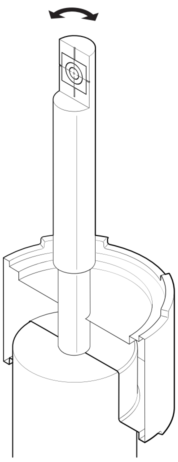
To measure up on top of a fixed mount GNSS receiver:

Steps	Description
	Mount the base on top of the GNSS receiver or antenna radome you are going to measure. The base should fit securely onto the features of the radome. If the base does not engage the radome features securely, flip the base over, and try to mount it again.

Steps	Description
	Remove the paper backing from the 25 mm acrylic target, exposing the adhesive. Place the target onto the rod. Smooth out the acrylic target to ensure good adhesion. **NOTE –** The 60 mm acrylic target is not used.
	Insert the target rod into the base. Rotate the rod so the acrylic target faces the expected position of the total station.

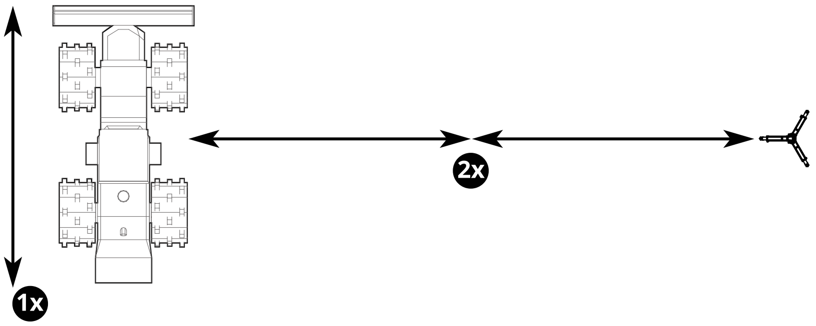
To measure up on the quick-release mast:

Steps	Description
	Mount the base on top of the mast. The base should fit securely. If not, flip the base over and try to mount it again.

Steps	Description
	Remove the paper backing from the 25 mm acrylic target, exposing the adhesive. Place the target onto the rod. Smooth out the acrylic target to ensure good adhesion. **NOTE –** The 60 mm acrylic target is not used.
	Insert the target rod into the base. The rod must be approximately vertical. Rotate the rod to face the expected position of the instrument.
	Repeat this process for the second mast if necessary.

2.4 Position the total station instrument

- Place the tripod on the right-hand side of the machine. You may perform the measure-up from the left, but the right side is recommended.
- Position the instrument approximately two machine lengths from the machine, ensuring that you can see all of the points through the instrument. If any targets are hidden, move the instrument.
- Ensure that the maximum horizontal angle between any acrylic target being measured and the total station is less than approximately 30°.



2.5 Measure-up

2.5.1 Overview

This section provides a general outline that enables technicians experienced with Trimble SCS900/Siteworks, or the total station's on-board software, to use a total station instrument to perform a compactor measure-up.

NOTE –Make sure the instrument you are using is within adjustment. Instrument adjustment can be performed on some models with SCS900/Siteworks if required, or by a SITECH® service center. An instrument that is not within adjustment may not provide accurate measurement results.

TIP – Use a USB flash drive to transfer measured points from the total station onto the system.

To use this method:

- To enable checking, before you begin the measure-up, make sure that the data collector is set to use the same measurement units and the same coordinate order as set in the Web Interface.
- Name the measured points as they are named in the Measure-up section of the Install Assistant.

For more information, see Using a USB flash drive to transfer measure-up data in .csv format.

2.5.2 Measurement workflow

1. Establish a position for the instrument.

   TIP – Use short coordinates to avoid having to enter extra digits in the Measure-up section in the Web Interface. For example, use Northing = 50, Easting = 30, Elevation = 10 for the instrument position.

2. Measure all measurement points.

   NOTE –The Target Height for the A measurements must be *0.00*.

3. Enter the measured coordinate values for each point into the fields in the Enter Coordinates screens in the Web Interface's Configure > Install Assistant > Measure-up section.

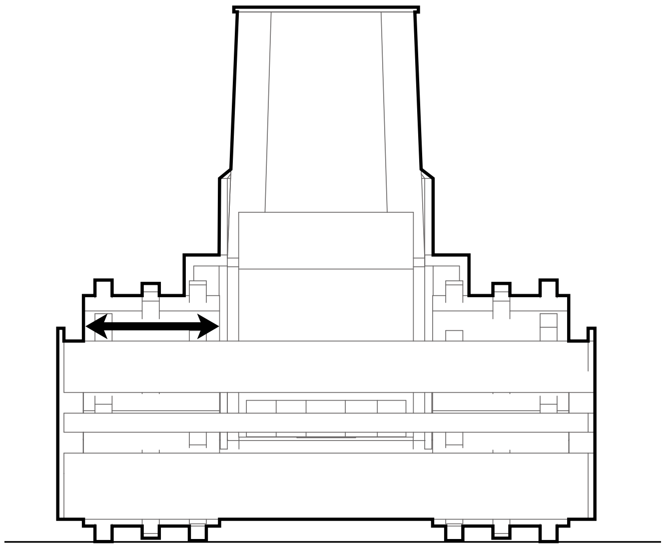
WARNING –Moving a machine with a GNSS measure-up tool attached can dislodge the tool from the machine. Always remove and store removable measure-up tools prior to moving the machine. Only move the machine when instructed in the Measure-up section. When the measured coordinate values are entered, the machine must be in the same position it was in when the measurements were taken.

2.5.3 Measure the drum

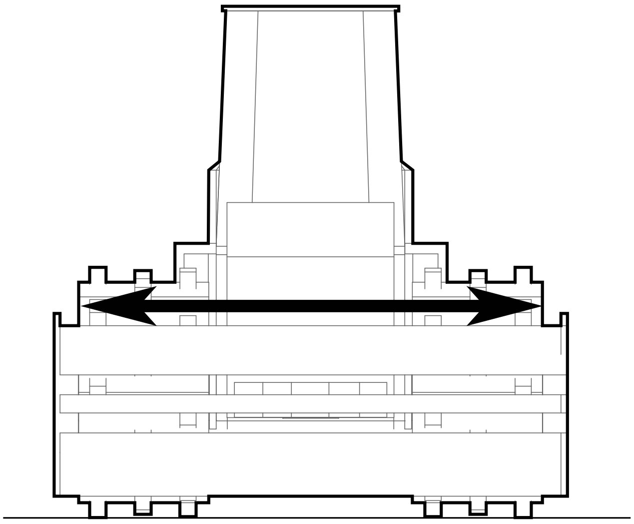
Use a tape measure to measure the width of the drum.

The landfill compactor requires 2 measurements:

- Drum Width – the width of one individual drum:



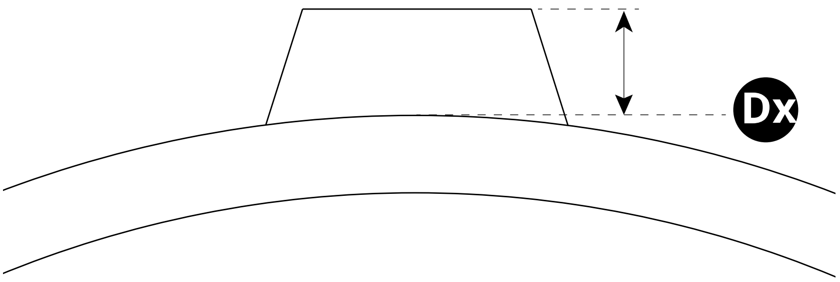
- Drum-to-Drum Width – the full width from the outside edge of the left drum to the outside edge of the right drum:



2.5.4 Measure the padfoot height (if applicable)

If the machine is equipped with a padfoot drum, use a tape measure to measure the height from the drum to the top of a padfoot. Measure from the surface of the drum, as marked by points D1, D2 and D3.

The padfoot height contributes to the calculations so ensure the measurement is accurate.



2.5.5 Entering the values into the Measure-up section in the Web Interface

Consider the following tips:

- Set your coordinate order in the Web Interface's Units dialog first to match your measured point coordinate order in SCS900/Siteworks.
- If using your laptop to enter the measured coordinates, you can select E, N, Elv at the same time for each measured point in a CSV file exported from SCS900/Siteworks and paste all three at the same time into the first field for each point in the Web Interface Measure-up sequences. This can increase the accuracy of data entry and decrease the time taken.

2.6 Using a USB flash drive to transfer measure-up data in .csv format

You can export comma separated value (.csv) files from most common survey data collectors onto a USB flash drive. You can edit .csv files using a text editor, or by opening the .csv file in a product such as Microsoft® Excel.

> **TIP –** For technicians working in regions where it is common for .csv files to use semicolons to separate fields, and commas as decimal fraction markers, then the Web Interface .csv import tool is also able to import these .csv files.

When formatted correctly, as described below, the .csv file can be imported into the Web Interface from the USB flash drive, reducing the amount of manual data entry required during machine commissioning.

2.6.1 Prerequisites

1. The measurements were made using a data collector running software capable of exporting .csv files.
2. The measurement units configured in the data collector are the same as the measurement units used in the Measure-up section of the Install Assistant.
3. The measured points are named as shown in the Measure-up section of the Install Assistant.

2.6.2 Exporting the machine measurements as a .csv file

On the data collector used to make the machine measurements, export the measured points to a USB flash drive in the usual way. If you are using Trimble SCS900 or Trimble Siteworks software on your data collector, make sure you select Include QA Data in the Export dialog, and select Memory Card as the destination.

2.6.3 If required, edit the .csv file

The final .csv file should resemble the example below when it is opened in a spreadsheet editor. This example shows a typical set of measurements.

The column headings must be:

- In English.
- Spelled as shown.
- Capitalized as shown.

> **NOTE –**The order of the columns and point measurements is not important. You can change the coordinate order for your project to suit the region, and you can measure the points in any order.

If you are editing the .csv file in a text editor, then the contents can look like:

```
Point Name,Northing,Easting,Elevation
```

If you are editing a semicolon separated .csv file in a text editor, then the contents can look like:

```
Point Name;Northing;Easting;Elevation
```

When you save the edited file, you must save it as a .csv file.

2.6.4 Import the .csv file into the Web Interface

1. Insert the USB flash drive into the device running the Web Interface.
2. Proceed through the Web Interface's Install Assistant > Measure-up section as usual until you reach the .csv file upload screen.

   TIP – If you choose the Skip to Entry option when it is offered, you will still have an opportunity to upload your .csv file.

3. In the .csv file upload screen, browse to the required file.
4. Select the Upload button. The file is uploaded and the measured points in the .csv file are copied into the Web Interface.
5. Proceed through the subsequent screens and confirm that the point data has been correctly imported.

NOTE – After measuring the points, the machine ***must not be moved*** until the .csv file import completes, and all the screens in the Measure-up section have been completed.

3

Landfill compactor measure-up and sensor calibration validation guide

In this guide:

3.1 Overview of compactor measure-up and sensor calibration validation

3.1.1 Purpose of this guide

This guide supports the validation of the compactor measure-up and sensor calibration as part of the commissioning of a new system.

The validation procedures enable you to:

- 3.2 Check horizontal guidance on the level
- 3.3 Check horizontal guidance on a slope

You can perform the checks in any order.

If the required accuracy is not achieved, the procedures describe diagnostic methods for identifying issues.

3.1.2 Prerequisites

Before starting the compactor measure-up and sensor calibration validation:

- Install the cab components and GNSS receiver, and validate the physical installation.
- Complete all the sections in the Web Interface's Configure > Install Assistant screen.
- In the Operator App's Work Settings > Drum Manager screen, verify the drum dimensions are correct. Update the Elevation Mapping Adjustment if required for the height of the padfoot.
- Know how to set-up and use a total station or a GNSS rover, and data collector.
- Validate the total station or GNSS rover on a known control point.

- If you are using a GNSS rover, make sure both the Earthworks system and the rover are using the same site calibration / coordinate system.
- A site with:
  - A level pad wide enough to perform 2 parallel passes with the compactor
  - A sloping pad with a cross slope representative of the slope the machine will work on
  - A swath of land that will compact when driven over by the compactor
- Review this entire guide. This ensures that you understand the full validation workflow.

3.1.3 Equipment list

- A total station in good adjustment or a GNSS receiver, with a fully charged battery
- If required, a data collector with a fully charged battery
- A note book and pens
- These instructions

3.2 Check horizontal guidance on the level

3.2.1 Check

1. Move the machine on to a level pad. Remove any articulation from the machine.
2. Set the drum focus to left.
3. Drive in a straight line for approximately 5 m (16 feet).
4. Ensure that the machine travel direction (forward/back) is correctly established and shown on the shortcut button in the Operator App.
5. Stop the machine and record the Northing, Easting, and Elevation values for the left drum side, either from the Operator App's Text Items or the Web Interface's Monitor > Machine Positions screen.
6. Use spray paint to mark a "T" on the ground on the drum's left side, comprising:
   - A line along the side of the drum.
   - A line marking the centerline of the drum: the axis along the ground beneath the axle of the drum.
7. Move the machine.
8. Use a rover (total station or GNSS) to record the position on the "T" and compare the readings. The readings should be within ±0.03 m (~1 inch).

3.2.2 Corrective action

If the drum position does not align with the starting position:

1. Ensure that the machine is not articulated.
2. Repeat the Measure-up section of the Web Interface's Configure > Install Assistant screen.

3.3 Check horizontal guidance on a slope

3.3.1 Check

Repeat the steps from Check horizontal guidance on the level using sloping ground.

3.3.2 Corrective action

If the drum position does not align with the starting position:

1. Ensure that the machine is not articulated.
2. Repeat the Measure-up section of the Web Interface's Configure > Install Assistant screen.

3.4 Validating the LCA settings

LCA (Landfill Compaction Algorithm) enables you to set the limit of what constitutes an optimally compacted material. In LCA mode, a pass is only counted when 2 drums pass over an area (usually the front and rear drum).

The system judges that the material is optimally compacted when the machine performs a pass and the material compacts within a defined limit. If the area continues to compact by more than the limit, more passes are required. An area that is optimally compacted appears green on the screen.

You can define several settings for the LCA:

- Expected Pass Count – The number of passes expected to complete optimum compaction on an area.
- Optimum Compaction Range (Max and Min)– If the elevation change after a pass is minimal (between the Min and the Max values), compaction is optimized and no additional passes are required.
- New Material Threshold – If elevation increases by more than this value, but less than the Thick Layer Threshold, it means new material has been added and the current pass count reverts to 1.
- Thick Layer Threshold – When new material is added and the increase in elevation is greater than this value, the layer is too thick and mapping appears black until material is removed and you tap the Reset Thick Layer shortcut key to reset the thick layer.

3.4.1 Prerequisite

Complete the installation process as described in the *Compactor Hardware Install Manual*.

Enter test values for the Landfill Compaction Algorithm section on the Web Interface's Mapping screen:

- Expected Pass Count: 8
- Optimum Compaction Range - Max: 100mm (4 inches)

- Optimum Compaction Range - Min: -100mm (-4 inches)
- New Material Threshold: 200mm (8 inches)
- Thick Layer Threshold: 600mm (24 inches)

On the Operator App, set the Mapping view to LCA.

3.4.2 Check

Validate the Optimum Compaction Range

1. Perform a pass on firm ground. The LCA view will display the pass as yellow.
2. Return over the ground. If the compaction is less than 100mm (4 inches), the pass will be in the Optimum Compaction Range and the LCA view will turn green.

Validate the New Material Threshold

1. Perform a pass on uncompacted material and then return over the material to the starting point. If each pass compacted the material by more than 100mm (4 inches), the LCA view should show:
   - First pass: yellow mapping (1 pass)
   - Return pass: pink mapping (2 passes)
2. Add 300mm (12 inches) of solid material to the compacted area.
3. Perform a single pass. The new material should exceed the New Material Threshold and the LCA view should revert to showing the pass as yellow (1 pass).

Validate the Thick Layer Threshold

1. Perform a pass with the compactor over uncompacted material and then return over the material to the starting point.
2. Add 1m (40 inches) of material to the compacted area.
3. Perform another pass. The new material should exceed the Thick Layer Threshold, causing the mapping to appear black.
4. Tap the Operator App's Reset Thick Layer shortcut key .
5. Select Yes to reset the thick layer.

Adjust the LCA settings

Adjust the LCA settings in the Web Interface to suit your site and material and then save the settings. Recommended settings are:

Setting	Recommended Value
Expected Pass Count	4

Setting	Recommended Value
Optimum Compaction Range (based on machine weight class)	27,220 kg (60,000 lbs.): • Max 150 mm (6 in) • Min -65 mm (-2.5 in)
	40,830 kg (90,000 lbs.): • Max 150 mm (6 in) • Min -100 mm (-4 in)
	54,440 kg (120,000 lbs.): • Max 150 mm (6 in) • Min -150 mm (-6 in)
New Material Threshold (all size classes)	200 mm (8 in)
Thick Layer Threshold (based on machine weight class)	27,220 kg (60,000 lbs.): 600 mm (24 in)
	40,830 kg (90,000 lbs.): 760 mm (30 in)
	54,440 kg (120,000 lbs.): 915 mm (36 in)

3.4.3 Corrective action

If the system does not respond as expected, check that guidance is correct and that the thresholds are being reached.

3.5 Save and activate machine files

3.5.1 Save a machine file

The current settings save automatically in the system memory and persist over power cycles. Current settings are overwritten when you activate or restore a saved machine file. Saving the current settings to a machine file makes the settings available for activating or restoring as needed.

> **NOTE –**Users with Operator Plus accounts can access the Machine Settings Files screen to apply machine files. From within the Operator App, tap System Settings > Manage Machine File. To give an operator Operator Plus access, within the Web Interface open Configure > Install Assistant > User Permissions.

Consider when to save a machine file:

- After installing, commissioning and validating guidance accuracy of the system
- Before and after a new valve calibration so you can compare previous settings to the new settings
- After completing and validating a measure-up (delete the previous machine file to avoid using the old settings instead of the new settings)

- After completing a new machine setup. You must complete a new valve air purge followed by a valve calibration and any other required machine calibrations for each machine using additional attachments and sensors. The machine file is then available for activation when the machine setup changes.

To save a machine file:

1. Open the Machine Settings Files page in the Web Interface.
2. Select ⋮ and then Save Settings.
3. Give the file a unique name and save.

3.5.2 Activate, or restore, a machine file

Activating, or restoring, a machine file is useful for returning the system back to a state that was previously saved.

Most system settings are overwritten with the values from the machine file when that machine file is activated or restored.

After activating a machine file, check the following items to ensure accurate guidance:

- Pad foot wear
- Currently loaded project and 2D bench or 3D design settings
- Operational settings such as focus
- Items such as Wi-Fi configurations and delayed shutdowns do not reset when activating a machine file.
- System licenses are not affected by activating a machine settings file.

To activate a machine setting file:

1. Open the Web Interface and then Advanced > Machine Settings.
2. Select the file to activate.
3. Select Make Active.

Notices

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